

Excluding All Irrational Mechanical Collatz Itineraries in Lean

Collatz proof project

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Abstract

We connect the repository's parity factor-complexity obstruction to its mechanical-word definition. Every mechanical parity itinerary has at most $L + 1$ distinct length- L factors, hence its natural Collatz orbit is bounded. A return of period q forces $q\alpha$ to be an integer. Consequently no natural seed, including any shifted seed, has a mechanical itinerary of irrational slope, for any real intercept. The proof is checked in Lean without added mathematical axioms. It closes a formal coverage gap between previous slope-specific results and a full irrational mechanical class exclusion. It does not prove Collatz; no mathematical priority is claimed.

1 Definitions and statement

Let $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n . Write $x_t = T^t(N)$ and let $p_N(L)$ count all distinct factors $(x_t \bmod 2, \dots, x_{t+L-1} \bmod 2)$ for $t \geq 0$. For real α, ρ , define

$$w_t = \lfloor (t+1)\alpha + \rho \rfloor - \lfloor t\alpha + \rho \rfloor.$$

The Lean predicate `HasItin` equates these integer letters with the actual parity values for every natural time.

Theorem 1. *For every irrational real α , real ρ , and natural N , the parity itinerary of N is not w . The same holds for every tail $T^s(N)$.*

No critical-density, continued-fraction, or positive-seed hypothesis is needed. The complexity implication below even allows arbitrary real slopes; the itinerary hypothesis itself enforces binary letters.

2 The ordered-threshold argument

Normalize a starting time t to $\beta_t = \{t\alpha + \rho\} \in [0, 1)$. Subtracting the integer part shows that the length- L factor is determined by $\lfloor i\alpha + \beta_t \rfloor$ for $0 \leq i \leq L$. For $1 \leq i \leq L$, put

$$d_i(\beta) = \lfloor i\alpha + \beta \rfloor - \lfloor i\alpha \rfloor, \quad R(\beta) = \sum_{i=1}^L d_i(\beta).$$

Each d_i is nondecreasing and takes values in $\{0, 1\}$, so $0 \leq R(\beta) \leq L$. If two intercepts have the same rank, order them. All coordinate differences then have the same sign and sum to zero; thus every coordinate agrees. Their successive floor differences agree as well. Choose one realizing start for each factor and assign its rank. This is an injection into $\{0, \dots, L\}$, proving

$$p_N(L) \leq L + 1.$$

The finite set of globally realized factors and the choice of representatives are noncomputable definitions, not an orbit-enumeration algorithm.

3 Boundedness and the irrational contradiction

The previously verified factor floor states that an unbounded orbit satisfies

$$5q + 1 \leq p_N(3q + K) \quad \text{whenever } N + 1 \leq 2^K.$$

Combining it with $p_N(L) \leq L + 1$ contradicts sufficiently large q . Thus a mechanical itinerary forces boundedness. Pigeonhole gives repeated states, and shifting to the first repeated state gives a return of some positive period q .

Let p be the odd-step count of that period. Determinism and additivity imply that k periods contain kp odd steps. Mechanical telescoping yields

$$kp = \lfloor kq\alpha + \rho' \rfloor - \lfloor \rho' \rfloor, \quad -1 < k(q\alpha - p) < 1 \quad (k \geq 0).$$

If $q\alpha - p$ were nonzero, the Archimedean property would violate one of these bounds for a sufficiently large natural k . Therefore $q\alpha = p$, contradicting irrationality because $q > 0$.

4 Formal declarations and verification

All new declarations are in `Collatz/MechanicalComplexity.lean`:

- `mechanical_complexity_le`: the global factor upper bound;
- `bounded_of_mechanical`: boundedness for any mechanical itinerary;
- `mechanical_return_slope`: integral slope times return period;
- `no_irrational_mechanical`: the full irrational-slope exclusion;
- `no_eventual_irrational_mechanical`: every tail is excluded.

The module builds under the repository’s Lean toolchain. These five statements are included in the selected axiom audit; their dependencies are confined to the standard foundations `propext`, `Classical.choice`, and `Quot.sound`. No external Sturmian equivalence or literature axiom is imported. This is a Lean build and axiom inspection, not an independent kernel replay.

5 Context and remaining problem

Dubickas proved a stronger asymptotic complexity lower bound for escaping Collatz trajectories in Theorem 5 of *On integer sequences generated by linear maps*, Glasgow Mathematical Journal 51 (2009), 243–252, [doi:10.1017/S0017089508004655](https://doi.org/10.1017/S0017089508004655). Our contribution here is the explicit formal bridge to mechanical words, using the repository’s weaker finite rational complexity floor.

This theorem subsumes the earlier irrational slope-family exclusions at the level of conclusions, while preserving their different quantitative proofs. It gives neither a complexity upper bound for arbitrary Collatz orbits nor a classification of all possible counterexamples. For rational mechanical slopes it proves boundedness but does not identify the eventual cycle. General escape and nontrivial cycles remain unresolved.